CYBER CRIME & DIGITAL FORENSICS PLATFORM

Terminal Setup, Dependency Installation & Live Evaluator Script

ACADEMIC LAB GUIDE

Target Host: 127.0.0.1

Official GitHub Repository: https://github.com/XxMySelfNooBxX/OS-Cyber
This repository contains the complete 3-tier architecture: Flask QA Target, FastAPI Orchestrator, React Dashboard UI, and custom Nuclei v3 compliance templates.

1. System Prerequisites Checklist

Before executing commands, ensure your friend's laptop has the following standard tools installed and added to the Windows PATH:

Prerequisite	Required Version	Verification Command	Purpose in Architecture
Python	3.11 or 3.12	python --version	Powers Flask target API (5000) & FastAPI orchestrator (8000)
Node.js / npm	Node 18+ / npm 9+	node -v; npm -v	Builds & serves React Vite Governance Dashboard (5173)
Git	2.30+	git --version	Clones repository and manages code versions
Nuclei Scanner	v3.11+	nuclei -version	Automated compliance scanner executing custom YAML templates

2. Quick One-Time Setup (Terminal Commands)

Open the **Antigravity Terminal** (or PowerShell as Administrator) and paste the following sequential block to clone the codebase and install all dependencies:

```
# Step A: Clone repository and enter directory
git clone https://github.com/XxMySelfNooBxX/OS-Cyber.git
cd OS-Cyber

# Step B: Install Python backend dependencies (Target App & Orchestrator)
python -m pip install -r target-app/requirements.txt
python -m pip install -r orchestrator-api/requirements.txt

# Step C: Install React Dashboard frontend dependencies
cd dashboard-ui
npm install
cd ..

# Step D: (Optional) Install Nuclei CLI if not present on system PATH
winget install projectdiscovery.nuclei
```

3. Launching the 3-Tier Platform (One-Click)

A portable startup script (**start_project.ps1**) is included in the project root. It dynamically locates Python on any machine and automatically spawns all three tiers:

```
powershell -ExecutionPolicy Bypass -File .\start_project.ps1
```

Tier Component	Local Binding URL	Backing Technology & Role
Tier 1: Target Flask App	http://127.0.0.1:5000	Simulated banking API; seeds cyber_bank.db with customer records and forensic logs
Tier 2: Orchestrator API	http://127.0.0.1:8000	FastAPI backend; manages forensics_vault.db, executes Nuclei scans, hashes evider
Tier 3: React Dashboard	http://127.0.0.1:5173	Dark-mode Cyber Governance Center; real-time compliance toggles, evidence inspec

4. Alternative Manual Launch (3 Separate Terminals)

If your friend prefers to view server logs in separate terminal windows, execute each command in its own PowerShell tab:

```
# --- TERMINAL 1: Target Flask Server (Port 5000) ---
cd target-app
python seed_db.py
python app.py

# --- TERMINAL 2: FastAPI Orchestrator (Port 8000) ---
cd orchestrator-api
python main.py

# --- TERMINAL 3: React Governance UI (Port 5173) ---
cd dashboard-ui
npm run dev
```

5. Step-by-Step Live Evaluator Demonstration Script

Have your friend follow this exact sequence when demonstrating the project to evaluators, professors, or exam committees:

Step 1: Open the Governance Dashboard

Open browser at **http://127.0.0.1:5173**. Point out the live connectivity indicator to the target server and the active posture badge: **RELAXED MODE — UNPROTECTED**.

Step 2: Execute Vulnerability Audit in Relaxed Mode

Click **'Run Live Nuclei Audit'**. Point out the live loading spinner while Nuclei scans the target API. The scan returns **4 Critical Vulnerabilities** (2 High, 1 Medium, 1 Low).

Step 3: Demonstrate Digital Forensic Evidence & Data Exfiltration

Inspect the findings: (a) **IDOR / PII Leak**: Live extraction of Rahul Sharma and Priya Verma's **PAN Card** & **Bank Account Numbers** without authentication. (b) **Debug Exposure**: Internal environment settings leaked. (c) **Security Headers Missing**: Absence of X-Frame-Options and CSP.

Step 4: Highlight Chain-of-Custody Hashing & Legal Compliance

Show the **SHA-256 Digital Forensics Integrity Hash** computed over the evidence payload. Click *'Copy Hash'* to prove tamper-evidence. Expand the **Legal & Regulatory Accordion** to show statutory mapping to **Section 43A of the Indian IT Act 2000** (body corporate data protection failure) and **DPDP Act 2023**.

Step 5: Demonstrate Real-Time Defensive Patching

Click **'Toggle Compliance Mode'**. The badge updates dynamically to **STRICT COMPLIANCE MODE — HARDENED**. Click **'Run Live Nuclei Audit'** again. The dashboard immediately displays the green banner: **'All Compliance Controls Verified — 0 Exposures Detected'**. Explain that authentication (HTTP 401), debug lockouts (HTTP 403), and CSP headers successfully neutralized all attack vectors.

Step 6: Forensic Audit History & Report Export

Use the **Scan History Sidebar** to switch between the Relaxed (breached) and Strict (remediated) reports. Click **'Print / Export'** to show the clean printable evidence report formatted for courtroom or forensic archival.

6. Troubleshooting & Common Pitfalls

Issue / Symptom	Root Cause	Quick Resolution
Script execution disabled error	Windows ExecutionPolicy restriction	Run: Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
Nuclei: command not found	Nuclei CLI not on system PATH	Run: winget install projectdiscovery.nuclei (then reopen terminal)
Port 5000 / 8000 already in use	Previous python background task running	Run: Get-Process python* | Stop-Process -Force
npm install fails / takes too long	Cached npm modules or slow mirror	Run: npm cache clean --force; npm install in dashboard-ui/